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Problem 1

$$(a) \text{var}(M_n) = \text{var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{\text{var}(X)}{n} \leq 0.01$$

$$n \geq \frac{\text{var}(X)}{0.01} = \frac{1}{0.01} = 100$$

$$(b) P(|M_n - h| \leq 0.05) \geq 0.99 \Leftrightarrow P(|M_n - h| > 0.05) \leq 0.01$$

$$\text{Chebyshev's inequality: } P(|M_n - h| > 0.05) \leq \frac{\text{var}(M_n)}{0.05^2} = \frac{\text{var}(X)}{n \cdot 0.05^2} = \frac{1}{n \cdot 0.05^2} = \frac{400}{n}$$

$$\text{Let } \frac{400}{n} \leq 0.01 \Leftrightarrow n \geq 40000$$

$$(c) \text{ In Part (a) } \text{var}(X) \leq \left(\frac{2.0 - 1.4}{2}\right)^2 = 0.09 \text{ m}^2$$

$$n \geq \frac{\text{var}(X)}{0.01} \Rightarrow n \geq \frac{\max(\text{var}(X))}{0.01} = 9$$

$$(b) \frac{\text{var}(X)}{n \cdot 0.05^2} \leq 0.01 \Leftrightarrow n \geq 200^2 \text{var}(X)$$

$$\Leftrightarrow n \geq \max(\text{var}(X)) \cdot 200^2 = 6^2 = 3600$$

Problem 2.

Denote ~~Xn~~ that X_n is RV. $X_n = \begin{cases} 1, & \text{smoker} \\ 0, & \text{non-smoker} \end{cases}$

$$S_n = X_1 + X_2 + \dots + X_n, \quad M_n = S_n/n$$

$$\text{var}(M_n) = \frac{\text{var}(X)}{n} \leq \frac{1}{n} \cdot \left(\frac{1-0}{2}\right)^2 = \frac{1}{4n}$$

$$P(|M_n - f| \geq \varepsilon) \leq \frac{\text{var}(M_n)}{\varepsilon^2} = \frac{1}{4n\varepsilon^2} \leq \delta \Rightarrow n \geq \frac{1}{4\delta\varepsilon^2} \Leftrightarrow n_{\min} = \left\lceil \frac{1}{4\delta\varepsilon^2} \right\rceil$$

$$(a) \text{ When } \varepsilon \text{ changes to } \frac{\varepsilon}{2}, n_{\min} = \left\lceil \frac{1}{4\delta \cdot (\frac{\varepsilon}{2})^2} \right\rceil = \left\lceil \frac{1}{\delta\varepsilon^2} \right\rceil$$

it is around the original n times 4

$$(b) \text{ When } \delta \text{ changes to } \frac{\delta}{2}, n_{\min} = \left\lceil \frac{1}{4(\frac{\delta}{2})\varepsilon^2} \right\rceil = \left\lceil \frac{1}{2\delta\varepsilon^2} \right\rceil$$

it is around the original n times 2.

Problem 3

(a) According to the Markov inequality, denote that X is the relevant score.

$$P(X \geq 85) \leq \frac{E[X]}{85} = \frac{15}{17}$$

(b) $65 \leq X \leq 85 \Leftrightarrow |X - 75| \leq 10 \Leftrightarrow |X - E[X]| \leq 10$

$$P(|X - E[X]| \geq 10) \leq \frac{\text{var}(X)}{10^2} = \frac{25}{100} = \frac{1}{4}$$

$$\Leftrightarrow P(65 \leq X \leq 85) = P(|X - 75| \leq 10) = 1 - P(|X - 75| > 10) \geq \frac{3}{4}$$

(c) Denote that $M_n = \frac{X_1 + X_2 + \dots + X_n}{n}$

$$P(|M_n - 75| \leq 5) \geq 0.9 \Leftrightarrow P(|M_n - 75| > 5) \leq 0.1$$

Based on Chebyshev's inequality,

$$P(|M_n - 75| > 5) \leq \frac{\text{var}(M_n)}{5^2} = \frac{\text{var}(X)}{n \cdot 5^2} = \frac{25}{n \times 25} = \frac{1}{n}$$

Let $\frac{1}{n} \leq 0.1 \Leftrightarrow n \geq 10 \Leftrightarrow$ at least 10 students would have to take the exam.

$$\begin{aligned} M_X(s) &= \sum_{n=0}^{\lambda-1} \frac{e^{-\lambda} \lambda^n}{n!} + \sum_{n \geq \lambda} \frac{e^{-\lambda} \lambda^n}{n!} \\ &= \sum_{n=0}^{\lambda-1} \frac{e^{-\lambda} \lambda^n}{n!} + \sum_{\substack{n=\lambda+1 \\ n \geq \lambda}} \frac{e^{-\lambda} \lambda^n}{n!} + \sum_{n \geq \lambda} \frac{e^{-\lambda} \lambda^n}{n!} \end{aligned}$$

$$\begin{aligned} M_X(s) &= \sum_{n=0}^{\lambda-1} \frac{e^{-\lambda} (\lambda e^s)^n}{n!} + \sum_{\substack{n=\lambda+1 \\ n \geq \lambda}} \frac{e^{-\lambda} (\lambda e^s)^n}{n!} + \sum_{n \geq \lambda} \frac{e^{-\lambda} (\lambda e^s)^n}{n!} \\ &\xrightarrow{\lambda e^s = 1} \end{aligned}$$

$$M_X(s) = \int_{-\infty}^a f_X(x) e^{sx} dx + \int_a^{+\infty} f_X(x) e^{sx} dx \geq e^{sa} P(X \leq a) \quad (s \leq 0)$$

$$\Leftrightarrow P(X \leq a) \leq e^{-sa} M_X(s)$$

$$\Leftrightarrow P(X \leq \lambda) \leq e^{-s\lambda} M_X(s) = e^{-s\lambda} \cdot e^{\lambda(e^s-1)} = e^{-s\lambda + \lambda(e^s-1)}$$

$$\hat{s} = \ln\left(\frac{\lambda}{\lambda}\right), \text{ L.H.S.} = (e^s)^{-\lambda} \cdot e^{-\lambda} \cdot e^{\lambda(e^s)} = \left(\frac{\lambda}{\lambda}\right)^{\lambda} \cdot e^{-\lambda} \cdot e^{\lambda\left(\frac{\lambda}{\lambda}\right)} = \frac{e^{-\lambda} (e\lambda)^{\lambda}}{\lambda^{\lambda}} \geq \text{L.H.S. - Q.E.D.}$$